

REBECCA K. LEANE



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RESEARCH INTERESTS

Theoretical Particle and Astroparticle Physics, Physics Beyond the Standard Model
Expertise: Indirect Dark Matter Detection, New Dark Matter and Astroparticle Search
Strategies, Intersection of Particle Theory and Astrophysical Data for Dark Matter

CURRENT POSITION

SLAC National Accelerator Laboratory, Stanford University

SLAC Particle Theory Group

Staff Scientist

2024–Present

Associate Staff Scientist

2021–2024

Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)

Senior Member

2021–Present

POSTDOCTORAL APPOINTMENTS

SLAC National Accelerator Laboratory, Stanford University

SLAC Particle Theory Group

Postdoctoral Research Associate

2020–2021

Massachusetts Institute of Technology

MIT Center for Theoretical Physics

Postdoctoral Research Associate

2017–2020

EDUCATION

University of Melbourne

2013–2017

Centre of Excellence for Particle Physics at the Terascale (CoEPP)

Ph.D. in Theoretical Particle Physics

Thesis Advisor: Prof. Nicole Bell

University of Cambridge

2012–2013

Department of Applied Mathematics and Theoretical Physics (DAMTP)

Master of Advanced Study in Mathematics (Part III of the Mathematical Tripos)

Essay Advisor: Prof. Ben Allanach

Monash University

2008–2011

Bachelor of Science Advanced with Honours

Majors: Physics and Mathematics. High Distinction.

Thesis Advisor: Prof. Csaba Balazs

GRANTS AND AWARDS	Mottez Fellowship, Institut Henri Poincaré, Sorbonne University, France (Invited Visiting Professor)	2025
	KIPAC Innovation Grant, \$42,000 USD Leane and Macintosh	2022
	NASA Fermi Guest Investigator Program, Cycle 12, \$70,000 USD Leane and Slatyer	2019–2020
	Chancellor’s Prize for Excellence in the PhD Thesis, U. Melbourne	2018
	Dean’s Award for Excellence in Graduate Research, U. Melbourne	2018
	Royal Society of Victoria Young Scientist Research Prize	2016
	Science Abroad Travelling Scholarship, U. Melbourne	2016
	Laby Foundation Early Career Researcher Travel Scholarship	2015
	Australian Postgraduate Award (APA)	2012–2016
	Monash Jubilee Honours Scholarship, Monash U.	2011
	J. L. William Honours Scholarship, Monash U.	2011

HIGHLIGHTED PUBLICATIONS

- **R. K. Leane**
Search for Dark Matter Annihilation and Decay with H α Line Emission
Accepted in PRL, [arXiv:[2512.09019](#) [hep-ph]]
- **R. K. Leane**, J. Beacom
Sub-GeV Dark Matter Direct Detection with Neutrino Observatories
Phys. Rev. Lett. 135, 191003 (2025), [arXiv:[2503.09685](#) [hep-ph]]
- C. Blanco and **R. K. Leane**
Search for Dark Matter Ionization on the Night Side of Jupiter with Cassini
Phys. Rev. Lett. 132, 261002 (2024), [arXiv:[2312.06758](#) [hep-ph]]
PRL Editor’s Suggestion
- A. Das, N. Kurinsky, **R. K. Leane**
Dark Matter Induced Power in Quantum Devices
Phys. Rev. Lett. 132, 121801 (2024) [arXiv:[2210.09313](#) [hep-ph]]
PRL Editor’s Suggestion
- **R. K. Leane** and T. Linden
First Analysis of Jupiter in Gamma Rays and a New Search for Dark Matter
Phys. Rev. Lett. 131, 071001 (2023), [arXiv:[2104.02068](#) [astro-ph.HE]]
- **R. K. Leane** and J. Smirnov
Exoplanets as Sub-GeV Dark Matter Detectors
Phys. Rev. Lett. 126, 161101 (2021), [arXiv:[2010.00015](#) [hep-ph]]
PRL Editor’s Suggestion
- **R. K. Leane** and T. R. Slatyer
Spurious Point Source Signals in the Galactic Center Excess
Phys. Rev. Lett. 125, 121105 (2020), [arXiv:[2002.12370](#) [astro-ph.HE]]
- **R. K. Leane** and T. R. Slatyer
Dark Matter Strikes Back at the Galactic Center
Phys. Rev. Lett. 123 (2019) 241101, [arXiv:[1904.08430](#) [astro-ph.HE]]
PRL Editor’s Suggestion

Author lists are in alphabetical order (per convention in theoretical particle physics) unless marked with an asterisk. Our field does not usually publish in Nature or Science, top journal is PRL. Underlined indicates a supervised student or SLAC postdoc.

Journal Articles:

52. T. Kosynets, **R. K. Leane**, J. Smirnov 2026
Dark Matter Weather:
Probing Sub-GeV Interactions with Earth-Shielding Modulation
To be submitted, [arXiv:[2607.20620](#) [hep-ph]]
[0 citations]
51. L. Santos-Olmsted, **R. K. Leane***, C. Blanco, J. Beacom 2025
Sub-GeV Dark Matter Detection with Dark Rates in Liquid Scintillators
Phys. Rev. D 113, 103007 (2026), [arXiv:[2512.13779](#) [hep-ph]]
[1 citation]
50. **R. K. Leane** 2025
Search for Dark Matter Annihilation and Decay with $H\alpha$ Line Emission
Accepted in PRL, [arXiv:[2512.09019](#) [hep-ph]]
[1 citation]
49. C. Blanco, M. Moore, **R. K. Leane** and J. Tong 2025
Complementary Planetary Spectroscopy Probes of Dark Matter
Phys. Rev. D 113, 123063 (2026), [arXiv:[2508.00980](#) [hep-ph]]
[4 citations]
48. **R. K. Leane***, J. Beacom 2025
Sub-GeV Dark Matter Direct Detection with Neutrino Observatories
Phys. Rev. Lett. 135, 191003 (2025), [arXiv:[2503.09685](#) [hep-ph]]
[9 citations]
47. C. Blanco, M. Moore, **R. K. Leane** and J. Tong 2024
Search for Dark Matter Induced Airglow in Planetary Atmospheres
Phys. Rev. Lett. 136, 251001 (2026), [arXiv:[2408.15318](#) [hep-ph]]
[12 citations]
46. I. John, **R. K. Leane**, T. Linden 2024
Dark Branches of Immortal Stars at the Galactic Center
Phys. Rev. D 112, 023028 (2025), [arXiv:[2405.12267](#) [astro-ph.HE]]
[21 citations]
45. M. Benito, K. Karchev, **R. K. Leane**, S. Podar, J. Smirnov, R. Trotta 2024
Dark Matter Halo Parameters from Overheated Exoplanets
via Bayesian Hierarchical Inference
JCAP 07 (2024) 038, [arXiv:[2405.09578](#) [astro-ph.IM]]
[12 citations]

44. **R. K. Leane** and J. Tong 2024
Optimal Celestial Bodies for Dark Matter Detection
JCAP 12 (2024) 031, [arXiv:[2405.05312](#) [hep-ph]]
[26 citations]
43. J. Acevedo, **R. K. Leane**, A. Reilly 2024
Detecting Dark Kinetic Heating in Exoplanets and Brown Dwarfs
JHEP 07 (2025) 079, [arXiv:[2405.02393](#) [hep-ph]]
[15 citations]
42. A. Das, N. Kurinsky, **R. K. Leane** 2024
Transmon Qubit Constraints on Dark Matter-Nucleon Scattering
JHEP 07 (2024) 233, [arXiv:[2405.00112](#) [hep-ph]]
[22 citations]
41. C. Blanco and **R. K. Leane** 2023
Search for Dark Matter Ionization on the Night Side of Jupiter with Cassini
Phys. Rev. Lett. 132, 261002 (2024), [arXiv:[2312.06758](#) [hep-ph]]
Selected as PRL Editor's Suggestion
[23 citations]
40. I. John, **R. K. Leane**, T. Linden 2023
Dark Matter Scattering Constraints from
Observations of Stars Surrounding Sgr A*
Phys. Rev. D 109, 123041 (2024), [arXiv:[2311.00015](#) [hep-ph]]
[29 citations]
39. J. Acevedo, **R. K. Leane**, L. Santos-Olmsted 2023
Milky Way White Dwarfs as Sub-GeV to Multi-TeV Dark Matter Detectors
JCAP 03 (2024) 042, [arXiv:[2309.10843](#) [hep-ph]]
[32 citations]
38. **R. K. Leane** and J. Smirnov 2023
Dark Matter Capture in Celestial Objects:
Treatment Across Kinematic and Interaction Regimes
JCAP 12 (2023) 040, [arXiv:[2309.00669](#) [hep-ph]]
[36 citations]
37. J. Acevedo, **R. K. Leane**, J. Smirnov 2023
Evaporation Barrier for Dark Matter in Celestial Bodies
JCAP 04 (2024) 038, [arXiv:[2303.01516](#) [hep-ph]]
[32 citations]
36. A. Das, N. Kurinsky, **R. K. Leane** 2022
Dark Matter Induced Power in Quantum Devices
Phys. Rev. Lett. 132, 121801 (2024), [arXiv:[2210.09313](#) [hep-ph]]
Selected as PRL Editor's Suggestion
[47 citations]
35. **R. K. Leane** and J. Smirnov 2022
Floating Dark Matter in Celestial Bodies
JCAP 10 (2023) 057, [arXiv:[2209.09834](#) [hep-ph]]
[52 citations]

34. [M. Collier](#), [D. Croon](#), **R. K. Leane** 2022
Tidal Love Numbers of Novel and Admixed Celestial Objects
Phys. Rev. D 106, 123027 (2022), [[arXiv:2205.15337](#) [gr-qc]]
[53 citations]
33. **R. K. Leane** and [T. Linden](#) 2021
First Analysis of Jupiter in Gamma Rays and a New Search for Dark Matter
Phys. Rev. Lett. 131, 071001 (2023), [[arXiv:2104.02068](#) [astro-ph.HE]]
[68 citations]
32. **R. K. Leane**, [T. Linden](#), [P. Mukhopadhyay](#), [N. Toro](#) 2021
Celestial-Body Focused Dark Matter Annihilation Throughout the Galaxy
Phys. Rev. D 103, 075030 (2021), [[arXiv:2101.12213](#) [astro-ph.HE]]
[92 citations]
31. **R. K. Leane** and [J. Smirnov](#) 2020
Exoplanets as Sub-GeV Dark Matter Detectors
Phys. Rev. Lett. 126, 161101 (2021), [[arXiv:2010.00015](#) [hep-ph]]
Selected as PRL Editor's Suggestion
[105 citations]
30. [D. Croon](#), [G. Elor](#), **R. K. Leane**, [S. McDermott](#) 2020
Supernova Muons: New Constraints on Z' Bosons, Axions, and ALPs
JHEP 01 (2021) 107, [[arXiv:2006.13942](#) [hep-ph]]
[171 citations]
29. **R. K. Leane** and [T. R. Slatyer](#) 2020
Spurious Point Source Signals in the Galactic Center Excess
Phys. Rev. Lett. 125, 121105 (2020), [[arXiv:2002.12370](#) [astro-ph.HE]]
[73 citations]
28. **R. K. Leane** and [T. R. Slatyer](#) 2020
The Enigmatic Galactic Center Excess:
Spurious Point Sources and Signal Mismodeling
Phys. Rev. D 102, 063019 (2020), [[arXiv:2002.12371](#) [astro-ph.HE]]
[66 citations]
27. [D. Hooper](#), **R. K. Leane**, [Y. Tsai](#), [S. Wegsman](#), [S. Witte](#) 2019
A Systematic Study of Hidden Sector Dark Matter:
Application to the Gamma-Ray and Antiproton Excesses
JHEP 07 (2020) 163, [[arXiv:1912.08821](#) [hep-ph]]
[40 citations]
26. [J. Acevedo](#), [J. Bramante](#), **R. K. Leane**, [N. Raj](#) 2019
Cooking Pasta with Dark Matter:
Kinetic and Annihilation Heating of Neutron Star Crusts
JCAP 03 (2020) 038, [[arXiv:1911.06334](#) [hep-ph]]
[110 citations]
25. **R. K. Leane** and [T. R. Slatyer](#) 2019
Dark Matter Strikes Back at the Galactic Center
Phys. Rev. Lett. 123 (2019) 241101, [[arXiv:1904.08430](#) [astro-ph.HE]]
Selected as PRL Editor's Suggestion
[162 citations]

24. HAWC Collaboration and J.F. Beacom, **R. K. Leane**, T. Linden, K. C.Y. Ng, A. Peter, B. Zhou
Constraints on Spin-Dependent Dark Matter Scattering
with Long-Lived Mediators from TeV Observations of the Sun with HAWC
Phys. Rev. D 98 (2018), 123012, [arXiv:[1808.05624](#) [hep-ph]].
[90 citations] 2018
23. HAWC Collaboration and J.F. Beacom, **R. K. Leane**, T. Linden, K. C.Y. Ng, A. Peter, B. Zhou
First HAWC Observations of the Sun Constrain Steady TeV Gamma-Ray Emission
Phys. Rev. D 98 (2018), 123011, [arXiv:[1808.05620](#) [astro-ph.HE]]
[52 citations] 2018
22. **R. K. Leane***, T. R. Slatyer, J. F. Beacom, K. C.Y. Ng
GeV-Scale Thermal WIMPs: Not Even Slightly Dead
Phys. Rev. D 98 (2018), 023016 [arXiv:[1805.10305](#) [hep-ph]]
[233 citations] 2018
21. N. F. Bell, Y. Cai, J. B. Dent, **R. K. Leane**, T. J. Weiler
Enhancing Dark Matter Annihilation Rates with Dark Bremsstrahlung
Phys. Rev. D 96 (2017), 023011 [arXiv:[1705.01105](#) [hep-ph]]
[44 citations] 2017
20. **R. K. Leane***, K. C.Y. Ng, J. F. Beacom
Powerful Solar Signatures of Long-Lived Dark Mediators
Phys. Rev. D 95 (2017), 123016 [arXiv:[1703.04629](#) [astro-ph.HE]]
[131 citations] 2017
19. N. F. Bell, Y. Cai, **R. K. Leane**
Impact of Mass Generation for Spin-1 Mediator Simplified Models
JCAP 01 (2017) 039 [arXiv:[1610.03063](#) [hep-ph]]
[66 citations] 2016
18. N. F. Bell, Y. Cai, **R. K. Leane**
Dark Forces in the Sky: Signals from Z' and the Dark Higgs
JCAP 08 (2016) 001 [arXiv:[1605.09382](#) [hep-ph]]
[52 citations] 2016
17. N. F. Bell, Y. Cai, **R. K. Leane**
Mono- W Dark Matter Signals at the LHC: Simplified Model Analysis
JCAP 01 (2016) 051 [arXiv:[1512.00476](#) [hep-ph]]
[66 citations] 2015
16. N. F. Bell, Y. Cai, J. B. Dent, **R. K. Leane**, T. J. Weiler
Dark matter at the LHC: Effective field theories and gauge invariance
Phys. Rev. D 92, 053008 (2015) [arXiv:[1503.07874](#) [hep-ph]]
[86 citations] 2015
15. N. F. Bell, Y. Cai, **R. K. Leane**, A. D. Medina
Leptophilic dark matter with Z' interactions
Phys. Rev. D 90, 035027 (2014) [arXiv:[1407.3001](#) [hep-ph]]
[93 citations] 2014

Review Articles:

14. **R. K. Leane** 2020
Indirect Detection of Dark Matter in the Galaxy
[arXiv:[2006.00513](#) [hep-ph]]
[70 citations]

Invited Viewpoint Articles:

13. Y. Hochberg, Y. Kahn, **R. K. Leane**, S. Rajendran,
K. Van Tilburg, T-T. Yu, K. Zurek 2022
New Approaches to Dark Matter Detection
[Nature Reviews Physics](#) (2022).
[24 citations]

White Papers and Working Group Reports:

12. A. Drlica-Wagner, C. Prescod-Weinstein, H.-B. Yu et al (incl **R. K. Leane**) 2022
Report of the Topical Group on Cosmic Probes of Dark Matter for Snowmass 2021,
[arXiv:[2209.08215](#) [hep-ph]]
Role: Report based in part on my Cosmic Frontier contributions.
[40 citations]
11. J. Cooley, T. Lin, H. Lippincott, T. Slatyer, T. Yu et al (incl **R. K. Leane**) 2022
Report of the Topical Group on Particle Dark Matter for Snowmass 2021,
[arXiv:[2209.07426](#) [hep-ph]]
Role: Report based in part on my Cosmic Frontier contributions.
[120 citations]
10. D. Green, J. Ruderman, B. Safdi, J. Shelton et al (incl **R. K. Leane**) 2022
Snowmass Theory Frontier: Astrophysics and Cosmology,
[arXiv:[2209.06854](#) [hep-ph]]
Role: Report based in part on my Theory Frontier contributions.
[33 citations]
9. L. Winslow, et al (incl **R. K. Leane**), 2022
Axion Dark Matter, [arXiv:[2203.14923](#) [hep-ex]]
Role: Contributor to “Synergies with astrophysical searches” section.
[344 citations]
8. M Baryakhtar, R. Caputo, D. Croon, K. Perez, et al (incl **R. K. Leane**), 2022
Dark Matter In Extreme Astrophysical Environments, [arXiv:[2203.07984](#) [hep-ph]]
Role: Contributor to “Light Dark Matter” section.
[106 citations]

7. **R. K. Leane***, S. Shin, L. Yang, et al, 2022
Snowmass2021 Cosmic Frontier:
Puzzling excesses in dark matter searches and how to resolve them,
[arXiv:[2203.06859](#) [hep-ph]]
Role: Paper coordinator/editor, and leader of the Galactic Center Excess section.
[46 citations]
6. T. Aramaki, P. von Doetinchem, S. Profumo, et al (incl **R. K. Leane**), 2022
Snowmass2021 Cosmic Frontier:
The landscape of cosmic-ray and high-energy photon probes of particle dark matter,
[arXiv:[2203.06894](#) [hep-ex]]
Role: Contributor to “Photon Probes” section.
[37 citations]
5. P. Harding, S. Horiuchi, D. Walker, et al (incl **R. K. Leane**), 2022
Snowmass2021 Cosmic Frontier:
Synergies between dark matter searches and multiwavelength/multimessenger as-
trophysics, [arXiv:[2203.06781](#) [hep-ph]]
Role: Contributor to “Searches with Celestial Bodies” section.
[10 citations]
4. D. Carney, N. Raj, et al (incl **R. K. Leane**), 2022
Snowmass2021 Cosmic Frontier:
Ultra-heavy particle dark matter, [arXiv:[2203.06508](#) [hep-ph]]
Role: Contributor to “Indirect Detection” section.
[88 citations]
3. K. Boddy, M. Lisanti, S. McDermott, N. Rodd, 2022
C, Weniger, et al (incl **R. K. Leane**),
Astrophysical and Cosmological Probes of Dark Matter,
[arXiv:[2203.06380](#) [hep-ph]]
Role: Contributor to “X-ray and γ -ray Dark Matter Signatures” section.
[88 citations]
2. M. Un Nisa, J. F. Beacom, S. Y. BenZvi, 2019
R. K. Leane, T. Linden, K. C.Y. Ng, A. H.G. Peter, B. Zhou
The Sun at GeV-TeV Energies: A New Laboratory for Astroparticle Physics
Astro2020 Science White Paper (2019) [arXiv:[1903.06349](#) [astro-ph.HE]]
Role: Contributor to text, content based in part on my solar dark matter search papers.
[40 citations]
1. G. Brooijmans et al (incl **R. K. Leane**) 2012
Les Houches 2011: Physics at TeV Colliders New Physics Working Group Report
(2012) [arXiv:[1203.1488](#) [hep-ph]]
Role: Contributor to text; “DLHA: Dark Matter Les Houches Agreement”, is heavily based on my bachelor thesis.
[123 citations]

TALKS

Summary: 18 invited colloquia, 24 invited plenary talks, 68 invited seminars, 35 additional conference/workshop talks (incl. additional 28 invited). Invited talks across Australia, Austria, Brazil, Canada, Denmark, France, Germany, Guadeloupe, Italy, Japan, New Zealand, Portugal, Spain, Sweden, Switzerland, United Kingdom, and the US.

After PhD, average 20+ invited talks per year; declined invitations are not listed below. Note most talks were declined in 2024 due to pregnancy/birth of child.

Invited Plenary Talks and Colloquia:

121. Invited Colloquium, South Dakota School of Mines & Technology, USA	Jan 2026
120. Invited Colloquium, San Francisco State University, SF CA, USA	Oct 2025
119. Invited Plenary Talk, PPC2025, Deadwood SD, USA	Jun 2025
118. Invited Plenary Talk, Dark Matter and Neutrinos 2025, Paris, France	May 2025
117. Invited Plenary Talk, APS April, Sacramento CA, USA	Apr 2024
116. Invited Colloquium, Texas A&M University, College Station TX, USA	Nov 2023
115. Invited Colloquium, University of Nevada, Las Vegas NV, USA	Sep 2023
114. Invited Plenary Talk, Pheno 2023, Pittsburgh PA, USA	May 2023
113. Invited Colloquium, University of Toronto, Toronto, Canada	Jan 2023
112. Invited Plenary Talk, TeVPA 2022, Kingston ON, Canada	Aug 2022
111. Invited Plenary Talk, Identification of DM 2022, Vienna, Austria	Jul 2022
110. Invited Colloquium, Snowmass Cosmic Frontier Meeting, Online	Jun 2022
109. Invited Overview Talk, HEP/Astro Results Forum, Online	May 2022
108. Invited Colloquium, Munich-Area Physics (TUM/LMU/MPI), Germany	May 2022
107. Invited Colloquium, SLAC, Menlo Park CA, USA	Feb 2022
106. Invited Colloquium, Carleton University, Ottawa, Canada	Dec 2021
105. Invited Colloquium, CERN, Geneva, Switzerland	Oct 2021
104. Invited Plenary Talk, Dark Matter 2021, Santander, Spain	Sep 2021
103. Invited Colloquium, North Carolina State U., Raleigh NC, USA	Sep 2021
102. Invited Plenary Talk, EXPLORE workshop, Germany	Aug 2021
101. Invited Plenary Talk, Planck 2021, Durham, UK	Jun 2021
100. Invited Colloquium, Stockholm University, Stockholm, Sweden	May 2021
99. Invited Colloquium, University of Chicago KICP, Chicago IL, USA	May 2021
98. Invited Colloquium, University at Albany SUNY, Albany NY, USA	Mar 2021
97. Invited Colloquium, Northeastern University, Boston MA, USA	Feb 2021
96. Invited Plenary Talk, Light Dark World 2020, Sydney, Australia	Dec 2020
95. Invited Plenary Talk, Identification of DM 2020, Vienna, Austria	Jul 2020
94. Invited Plenary Talk, 3rd World Summit on EDSU, Guadeloupe	Mar 2020
93. Invited Colloquium, MIT, LNS, Cambridge MA, USA	Feb 2020
92. Invited Plenary Talk, DM@LHC 2019, Seattle WA, USA	Aug 2019

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| 91. Invited Colloquium, University of Melbourne, Melbourne Australia | Dec 2018 |
| 90. Invited Plenary Talk, CAASTRO-CoEPP Joint Workshop, Australia | Jan 2017 |
| 89. Invited Plenary Talk, ATLAS Astroparticle Forum Meeting, CERN | Nov 2016 |

Invited Seminars and Other Conference Talks:

- | | |
|---|----------|
| 88. Invited Seminar, Berkeley Center for Theoretical Phys, Berkeley, USA | Apr 2026 |
| 87. Invited Seminar, University of Hawaii, Honolulu HI, USA | Feb 2026 |
| 86. Invited Talk, SLAC LZ Collaboration Meeting, Menlo Park CA, USA | Feb 2026 |
| 85. Invited Talk, Search for New Physics Workshop, Salt Lake City UT, USA | Mar 2025 |
| 84. Invited Seminar, Perimeter Institute, Waterloo ON, Canada | Feb 2024 |
| 83. Invited Seminar, Latin-American Webinars in Physics, Online | Nov 2023 |
| 82. Invited Seminar, University of Oregon, Eugene OR, USA | Oct 2023 |
| 81. Invited Talk, Galactic Center Excess Workshop, Rutgers NY, USA | Jun 2023 |
| 80. Invited Talk, Future Gamma Telescope Workshop, Chicago IL, USA | May 2023 |
| 79. Invited Symposium Speaker, APS April Conference, Minnesota, USA | Apr 2023 |
| 78. Invited Talk, Aspen Winter Conference, Aspen CO, USA | Mar 2023 |
| 77. Invited Talk, Stony Brook Simons Workshop, Stony Brook NY, USA | Mar 2023 |
| 76. Invited Seminar, McGill University, Montreal, Québec, Canada | Nov 2022 |
| 75. Invited Seminar, Fermilab, Batavia IL, USA | Oct 2022 |
| 74. Invited Seminar, University of Utah, Salt Lake City UT, USA | Sep 2022 |
| 73. Invited Seminar, University of Hawaii, Manoa HI, USA | May 2022 |
| 72. Invited Seminar, Stanford University, Stanford CA, USA | Apr 2022 |
| 71. Invited Seminar, Kings College London, London, UK | Mar 2022 |
| 70. Invited Seminar, University of Kentucky, Lexington KY, USA | Mar 2022 |
| 69. Invited Talk, Bay Area Theoretical Physics Seminar, San Francisco CA | Mar 2022 |
| 68. Invited Talk, FISICA 2022, Mainz, Germany | Mar 2022 |
| 67. Invited Seminar, Rutgers University, Piscataway NJ, USA | Dec 2021 |
| 66. Invited Seminar, Colgate University, Hamilton NY, USA | Nov 2021 |
| 65. Invited Parallel Talk, PANIC 2021, Lisbon, Portugal | Sep 2021 |
| 64. Invited Seminar, Max-Planck Institute, Heidelberg, Germany | Jul 2021 |
| 63. Invited Seminar, UCLA, Los Angeles CA, USA | Jun 2021 |
| 62. Invited Seminar, Imperial College London, London, UK | Apr 2021 |
| 61. Invited Seminar, University of Torino, Torino, Italy | Apr 2021 |
| 60. Invited Talk, Aspen Center for Physics Winter Conf., Aspen CO, USA | Mar 2021 |
| 59. Invited Seminar, University of Notre Dame, Notre Dame IN, USA | Mar 2021 |
| 58. Invited Seminar, Brookhaven National Laboratory, Upton NY, USA | Feb 2021 |
| 57. Invited Seminar, Queen's University, Kingston ON, Canada | Feb 2021 |
| 56. Invited Seminar, Stanford University, Stanford CA, USA | Dec 2020 |

55. Invited Seminar, California Institute of Technology, Pasadena CA, USA Dec 2020
54. Invited Talk, 3rd South American DM Workshop, Brazil Dec 2020
53. Invited Seminar, UC Santa Cruz, Santa Cruz CA, USA Nov 2020
52. Invited Seminar, SLAC, Menlo Park CA, USA Nov 2020
51. Invited Talk, Snowmass CF-01 Group Meeting Aug 2020
50. Invited Seminar, Sydney CCPC, Sydney, Australia Aug 2020
49. Invited Seminar, Kavli IPMU, Tokyo, Japan May 2020
48. Parallel Talk, Pheno 2020, Pittsburgh PA, USA May 2020
47. Invited Talk, Astrophysical Signatures of DM Workshop, MI, USA May 2020
(*canceled due to covid-19*)
46. Invited Talk, Aspen Winter Conf. on Particle Physics, Aspen CO, USA Mar 2020
(*canceled due to covid-19*)
45. Invited Seminar, U. Mass Amherst, Amherst MA, USA Feb 2020
44. Invited Seminar, Tufts University, Medford MA, USA Feb 2020
43. Invited Seminar, Boston University, Boston MA, USA Nov 2019
42. Invited Seminar, SLAC, Menlo Park CA, USA Oct 2019
41. Invited Seminar, Texas A&M University, College Station TX, USA Oct 2019
40. Invited Seminar, MIT Center for Theoretical Physics, Camb. MA, USA Sep 2019
39. Invited Seminar, Perimeter Institute, Waterloo ON, Canada Jun 2019
38. Invited Talk, Aspen Center for Physics, Aspen CO, USA Jun 2019
37. Invited Seminar, KICP, University of Chicago, Chicago IL, USA Apr 2019
36. Invited Seminar, Brown University, Providence RI, USA Apr 2019
35. Invited Seminar, Fermilab, Batavia IL, USA Mar 2019
34. Invited Seminar, Brandeis University, Boston MA, USA Mar 2019
33. Invited Seminar, Princeton University, Princeton NJ, USA Feb 2019
32. Invited Talk, TRIUMF Dark Matter Workshop, Vancouver BC, Canada Feb 2019
31. Parallel Talk, DESY Theory Workshop, Hamburg, Germany Sep 2018
30. Invited Seminar, University of Melbourne, Melbourne VIC, Australia Sep 2018
29. Parallel Talk 1, IDM 2018, Providence RI, USA Jul 2018
28. Parallel Talk 2, IDM 2018, Providence RI, USA Jul 2018
27. Invited Parallel Talk, PASCOS 2018, Cleveland OH, USA Jun 2018
26. Parallel Talk, Pheno 2018 Symposium, Pittsburgh PA, USA May 2018
25. Invited Seminar, California Institute of Technology, Pasadena CA, USA Apr 2018
24. Invited Seminar, LBNL, Berkeley CA, USA Apr 2018
23. Invited Seminar, University of Michigan, Ann Arbor MI, USA Apr 2018
22. Invited Seminar, Dartmouth University, Hanover NH, USA Apr 2018
21. Invited Seminar, University of Melbourne, Melbourne VIC, Australia Jan 2018
20. Invited Seminar, MIT Center for Theoretical Physics, Camb. MA, USA Sep 2017

19. PhD Completion Seminar, University of Melbourne, Australia	Mar 2017
18. Invited Seminar, TRIUMF, Vancouver BC, Canada (remote)	Dec 2016
17. Invited Talk, CETUP Dark Matter Workshop, Deadwood SD, USA	Jul 2016
16. Invited Seminar, University of Cincinnati, Cincinnati OH, USA	May 2016
15. Invited Seminar, CCAPP, Ohio State University, Columbus OH, USA	Jun 2016
14. Invited Seminar, UC Riverside, Riverside CA, USA	May 2016
13. Invited Seminar, UC Irvine, Irvine CA, USA	May 2016
12. Parallel Talk, Pheno 2016 Symposium, Pittsburgh PA, USA	May 2016
11. Invited Seminar, University of Melbourne, Melbourne VIC, Australia	Apr 2016
10. Invited Seminar, SLAC, Menlo Park CA, USA	Oct 2015
9. Invited Seminar, CCAPP, Ohio State University, Columbus OH, USA	Oct 2015
8. Invited Seminar, Fermilab, Batavia IL, USA	Sep 2015
7. Invited Seminar, Vanderbilt University, Nashville TN, USA	Sep 2015
6. Invited Seminar, University of Melbourne, Melbourne VIC, Australia	Aug 2015
5. Parallel Talk, SUSY 2015, Lake Tahoe CA, USA	Aug 2015
4. Invited Seminar, University of Melbourne, Melbourne VIC, Australia	May 2015
3. Parallel Talk, CoSPA2014, Auckland, New Zealand	Dec 2014
2. Contributed Talk, RESCEU APCosPA School, Matsumoto, Japan	Jul 2014
1. Invited Seminar, University of Cambridge, Cambridge, UK	Jun 2013

STANFORD/SLAC SERVICE	SLAC Physics of the Universe Strategic Plan Steering Committee	2025–2026
	KIPAC Postdoc Selection Committee	2025
	Stanford Physics, Identity, and Equity Program Invited Speaker	2021, 2022, 2023, 2024
	SLAC Linac Coherent Light Source (LCLS) Proposal Reviewer	2024, 2025
	SLAC Panofsky Fellowship Search Committee	2022–23
	Stanford Physics Graduate Admissions, Astrophysics Committee	2022–23
	Dark Matter Topical Lead for DOE HEP Institutional Review of SLAC	2022
	SLAC Particle Theory Graduate Spring Open House Co-ordinator	2022–24
	Stanford Selection Committee for CAMPARE/Leadership Alliance Programs	2022
	Stanford Physics Graduate Admissions, Astrophysics Committee	2021–22
	Stanford Physics Recruitment Committee	2021–22
	Stanford Graduate Student Orientation: SLAC Theoretical Physics Overview Talk	2021
	SLAC Particle Theory Postdoc Selection Committee	2021–
	SLAC Particle Theory Website Maintenance	2021–22
	SLAC Dark Matter Meetings/Journal Club Founding Organizer	2021–
	Stanford Physics PhD Thesis Committee / PhD Defense Examiner:	
	• Yawen Xiao (advisor: Peter Graham)	2025–2026
	• Samuel Wong (advisor: Peter Graham)	2023–2025
	• David Cyncynates (advisor: Savas Dimopoulos)	2022

MENTORING & SUPERVISING	SLAC Postdoctoral Researchers:	
	Linda Xu	2024–
	Javier Acevedo (→ Postdoc at U. Victoria/TRIUMF)	2022–2025
	Anirban Das (→ Faculty at Saha Institute)	2021–2022
	PhD Students:	
	<i>Stanford Thesis Advisor:</i>	
	Lillian Santos-Olmsted	2023–
	Joshua Tong	2022–2026
	<i>Supervising other Stanford PhD student projects:</i>	
	Gowri Sundaresan	2025–
	Aidan Reilly (→ Postdoc at Stony Brook, YITP)	2023–2024
	Payel Mukhopadhyay (→ Postdoc at Berkeley → Faculty at U. Cambridge)	2020–2021
	<i>Supervising external PhD student projects:</i>	
	Ian Holst, U. Chicago (DOE SCGSR fellowship, hosted by Leane at SLAC) (→ Postdoc at U. Toronto)	2025–2026
	Thong Nguyen, Stockholm University	2025–
	Marianne Moore, MIT (→ Postdoc at Fermilab/U. Chicago)	2024–2026
	Isabelle John, Stockholm University (→ Postdoc at U. Turin)	2023–2024
	Michael Collier, Durham University	2022
	Javier Acevedo, Queens University (→ Postdoc at SLAC)	2019
	Undergraduate Students:	
	Iz Paylor, Stanford	Summer 2022
	Afura Taylor, MIT (→ PhD at Princeton)	2020–2021
TEACHING	Invited Lecturer, Dark Matter Searches, Pre-SUSY Summer School, CA, USA	2025
	Invited Lecturer, Dark Matter in Astrophysics Lectures, N3AS Summer School, USA	2024
	Invited Lecturer, Dark Matter Theory Lectures, SLAC Summer Institute, USA	2022
	Invited Lecturer, EXPLORE2021 Summer School Workshop, Germany	2021
	Guest Lecturer for MIT graduate course <i>8.811: Particle Physics II</i>	2017
	ATLAS International Masterclass Tutor, U. Melbourne	2015
	Physics Teaching Assistant: U. Melbourne, Monash U.	2011–2015

EXTERNAL REVIEWING	Editorial Boards / Journal Editor:	
	• Journal of Cosmology and Astroparticle Physics (JCAP)	2026–
	Institution/Personnel Reviewing:	2021–
	• US Department of Energy (DOE) University Comparative Review Panel Theory Frontier, Energy Frontier	Multiple years
	• Tenure review cases worldwide	
	Grant Reviewing:	2018–
	• NASA Hubble Fellowship Selection Panel, Physics & Cosmology, USA	
	• NASA Astrophysics Data Analysis Program, USA	
	• US Department of Energy (DOE), USA	
	• Natural Sciences and Engineering Research Council of Canada (NSERC), Canada	
	• Marsden Fund Council, New Zealand	
	• National Science Centre, Poland	
	• MIT Heising-Simons Physics Research Fellowship Review Panel, USA	
	Journal Referee:	2018–
	• Physical Review Journals (PRL, PRD)	
OTHER COMMUNITY	• Journal of High Energy Physics (JHEP)	
	• Journal of Cosmology and Astroparticle Physics (JCAP)	
	• Physics Letters B (PLB)	
	Invited Viewpoint Article Contributor in Nature Physics Reviews, “New Approaches to Dark Matter Detection” (see publications)	2022
	Invited Coordinator, Editor, and Author	2021–2022
	Snowmass Cosmic Frontier CF01 White Paper: “Puzzling Excesses in Dark Matter Searches and How to Resolve Them”	
	Contributor to 6 Cosmic Frontier Snowmass White Papers (see publications)	2022
	Contributor to 1 Theory Frontier Snowmass White Paper (see publications)	2022
	Coordinator/Principal Author of Two Snowmass Letters of Interest	2020
	Understanding the Galactic Center Gamma-Ray Excess: Observational Prospects	
	Understanding the Galactic Center Gamma-Ray Excess: Theory Prospects	
	Submitted to Cosmic Frontier and Theory Frontier Snowmass Working Groups	
	Co-organizer/convener, Galactic Center Excess Workshop, Rutgers	2023
	Convener/Chair, Plenary Session, TeV Particle Astrophysics (TeVPA) Conference	2022
	SLAC Dark Matter Meetings/Journal Club Founding Organizer	2021–
PUBLIC TALKS & PANELS	Particle Theory Seminar Organizer, MIT Center for Theoretical Physics	2018–2020
	Convener, Dark Matter Session, TeV Particle Astrophysics (TeVPA) Conference	2019
	Public Lecture, Stanford University, Stanford CA, USA	2023
	Young Scientist Research Prize Talk, Royal Society of Victoria, Australia	2016
	Public Talk, Strathcona Girls’ School, Australia	2014
	Panel Member, National Science Week: Kids Ask Questions, Australia	2013
	Panel Member, National Science Week: Ask a Scientist, Australia	2013
	Public Talk, Monash University Open Day Seminar Series, Australia	2011

PRESS	Interview, <i>Phys.Org</i> ,	2025
INTERVIEWS	Study highlights the potential of neutrino observatories to search for light dark matter	
& ARTICLES	Press Piece, <i>Nature</i> ,	2025
	‘Immortal’ stars have an elixir of youth: dark matter	
	Press Piece, <i>Astronomy Magazine</i> ,	2024
	Scientists use Jupiter to search for dark matter	
	Interview, <i>SLAC News</i> ,	2023
	Scientists propose a new way to search for dark matter	
	Interview, <i>New Scientist</i> ,	2022
	Earth’s Surface May Be Teeming with Trillions of Dark Matter Particles	
	Video Interview (~ 10,000 views), <i>Universe Today</i> ,	2021
	Interview with Dr. Rebecca K. Leane, Detecting Dark Matter	
	Interview, <i>SLAC News</i> ,	2021
	How Exoplanets Could Aid the Search for Dark Matter	
	Interview, <i>Universe Today</i> ,	2021
	Jupiter Could Make an Ideal Dark Matter Detector	
	Press Piece, <i>APS Physics</i> ,	2021
	Detecting Dark Matter in Exoplanets	
	Interview, <i>Science Magazine</i> ,	2021
	Dark Matter Could Warm the Hearts of Lonely Old Planets, Scientists Predict	
	Interview, <i>Wired Magazine</i> ,	2021
	Where’s the Dark Matter? Look for Suspiciously Warm Planets	
	Interview, <i>Quanta Magazine</i> ,	2020
	Physicists are Expanding the Search for Dark Matter	
	Interview, <i>The Boston Globe Newspaper</i> ,	2019
	MIT scientists find DM could be cause of mysterious energy at the center of our galaxy	
	Interview, <i>MIT News</i> ,	2019
	Is there dark matter at the center of the Milky Way?	
	Press Piece, <i>APS Physics</i> ,	2019
	New Hope for Milky-Way Dark Matter	
	Interview, <i>Quanta Magazine</i> ,	2019
	Dark Matter Gets a Reprieve in New Analysis	
	Interview, <i>Kavli Foundation</i> ,	2019
	Dark Matter is Back	
	Interview, <i>Science Magazine</i> ,	2019
	Physicists Revive Hunt for Dark Matter in the Heart of the Milky Way	
	Interview, <i>Popular Mechanics</i> ,	2019
	Filling the Void: What is Dark Matter?	
	High impact papers on Altmetric (stats as of 06/14/21):	
	• “Exoplanets as Sub-GeV Dark Matter Detectors” (2021) paper w/ Juri Smirnov: Altmetric Score of 435, ranked #49 of all time for media coverage of PRL outputs • “Dark Matter Strikes Back at the Galactic Center” (2019) paper w/ Tracy Slatyer: Altmetric Score of 327, ranked #91 of all time for media coverage of PRL outputs	
FILM WRITING	Key Scientific Researcher on International Science Documentary Series “Uranium: Twisting the Dragon’s Tail”	2014–2015
	Worked with Emmy Award winning producer Sonya Pemberton and writer Wain Fimeri part time in a paid role over 12 months. Had a major role explaining the science content and developing the script for the series. Aired worldwide (on PBS in the US) and won numerous science communication and film awards.	